

Rapido: Accelerating ride-hailing journeys across India with optimized routes

Tell us your challenge. We're here to help.

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[About Rapido](#)

With Google Map
Mobility services
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About Rapido

Founded in 2015, Rapido operates mobile app-based taxi services in 100 cities in India.

Customers have taken more than 1 billion rides since Rapido's inception.

Industries: Transportation & Logistics

Location: India

[Contact us \(/contact\)](#)

Google Maps Platform results

- Slashed call center calls from riders from 5% to less than 1%
- Improved rider retention rates by enhancing their trust in the platform
- Supported diversification into offerings, such as shared car services

Improve
track
accuracy
about
about

search

About Searce

Founded in 2004, Searce is an engineering-led modern tech consultancy that empowers clients to futurify by delivering real business outcomes, leveraging Cloud, AI, Data & Analytics. With a dedicated Location Intelligence Practice, they won the 2023 Google Cloud Specialization Partner of the Year for Location-Based Services.

Services.

Google Maps Platform

(<https://mapsplatform.google.com/>)

Traffic congestion is a perennial problem in India's largest cities. Fortunately, bike taxis captained by skilled riders and managed by mobile apps can navigate bottlenecks by weaving in between vehicles to get people and packages promptly to their destinations.

Founded in 2015, Rapido (<https://www.rapido.bike/>) is a leading bike taxi operator that also offers three- and four-wheel taxi services to customers in 100 cities around India. Today, the business retains four million

drivers, known as Captains, who complete 45 million rides per month. In the next few years, the company has ambitious growth plans, including reaching 100 million monthly rides.

According to Srivatsa Katta, Head of Engineering at Rapido, capacity is not an obstacle to achieving this goal: About 12 million new two-wheelers, 100,000 three-wheelers, and 50,000 four-wheelers are coming to market every month.

"Despite customers taking more than one billion rides since Rapido's inception, we haven't yet tapped into more than 5% of the addressable two-wheeler market," explains Katta. "With Google Maps Platform

Mobility services

(<https://developers.google.com/maps/documentation/transportation-logistics/mobility>)

integrated into our mobile app, we can look beyond the thriving urban centers classified in India as 'tier one' cities to 'tier two' emerging urban centers and 'tier three' growing urban centers."

Mobility services give Rapido precise mapping and location solutions to help the business identify where its Captains are positioned, where they need to pick up and drop off a customer, and the best route to take.

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Engineering, Rapido*

"These services allow us to live up to our promises to customers about the cost of a trip, and our promises to Captains that they receive fair payment for the number of journeys they take and the distance they travel," says Katta.

Driving toward more accurate mapping and location services

Since its establishment, the taxi operator had relied on Google Maps (<https://maps.google.com/>) in combination with other mapping services for a range of business operations. This decision was largely determined by the degree of geolocation data precision required by a particular team or function.

However, to improve the utilization and earnings of its Captains, provide customers with accurate, timely information about their journeys, build trust with both cohorts, and position the business for growth, Rapido needed to embed accurate mapping and location services into its mobile app.

A key catalyst for change was to overcome the tendency of many Captains to switch off their GPS systems once they had started a journey, hampering the business's ability to track their adherence to a

specified route. Rapido needed a solution that enabled real-time tracking through a combination of data sources and signals, including GPS and sensors embedded in mobile devices.

Integrating location and route data into the Rapido app

With the APIs and SDKs available through Google Maps Platform and Mobility services, Rapido is integrating a range of functionality into its app. This includes route planning and dispatch that gives customers an accurate estimated time of arrival and the ability to track drivers' routes in real time. It will also provide route calculations, driver routing, and navigation for two-wheel vehicles, as well as live traffic information and address context and validation to ensure drivers go to the right place.

With route planning and dispatch, customers can plan appointments and engagements around forecast travel times, including the ETA of a Captain assigned to their booking, and have clear visibility of the route they are traveling. Prior to the integration of Mobility services, customers found themselves delayed by fast-changing traffic patterns or construction work that Rapido's previous map service had not updated or anticipated.

For Rapido, route planning and dispatch and address validation enables the business to adhere to travel times that customers take as promises on which to plan their days.

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Route planning and dispatch also enables Rapido to more effectively pair Captains with customers. "We don't simply rely on distance to determine this; a Captain may be closest to the customer but on the opposite side of the road in heavy traffic and unable to make a U-turn for several kilometers," says Katta.

"Navigating these nuances is where route planning and dispatch plays a significant role."

The business has improved its ability to accurately capture Captains' locations in real-time from as low as 60% to close to 95%, improving its ability to monitor and manage its fleet to optimize Captain and customer experiences.

With driver routing and navigation, Rapido is guiding its Captains in real time through the fast-changing traffic patterns of India's tier one cities. This helps them effectively bypass accidents, construction work, or other factors to optimize journey times and increase productivity, earnings, and rider retention rates. Meanwhile, address context and validation gives Captains and customers greater certainty that pickups will not be missed and that they're headed toward the correct destination.

These capabilities are helping entrench trust in the Rapido platform among internal and external stakeholders. "It is important to set correct price expectations for customers when they request rides, and accurate earnings expectations for Captains," explains Katta. "We can do that by being accurate about distances, routes, and timings in matching a Captain to a customer."

With Mobility services, Rapido is now adding new capabilities to its mobile app such as one-click bookings and safety features including alerts to customers and emergency contacts if a Captain deviates significantly from an agreed route. "These features demonstrate that we genuinely care about

our customers, further building trust in our platform," says Katta.

Turning to an experienced partner for help

The business needed skilled external expertise to help deliver the project and turned to Google Cloud Partner [Searce](https://www.searce.com/) (<https://www.searce.com/>) for help. Searce advised Rapido that Mobility services could add features, such as accurate, up-to-date traffic information, including predictive capabilities, into the app.

The fact that many Captains and customers used Google Maps outside the Rapido context also played a key role in the business's decision to select Mobility services. "As people in India were already familiar with the Google Maps interface, we could expect a smoother adoption of new features supported by Google Maps Platform in the Rapido app," explains Katta.

However, Rapido's engineering team needed to convince the business that a new mapping platform was worth the resource commitment and change management involved. "To make our case, we ran a limited-time proof of concept in two cities to showcase improvements in business metrics, such as the number of Captains' who call the Rapido call center about directions or similar operational issues, as a percentage of overall calls," says Katta. "This had fallen from 5% to less than 1%, meaning Captains had more time to take more customers, boost their earnings, and improve retention rates."

Collaboration between Searce and the Google Maps Platform team was integral to the success of the proof of concept, with the partner reporting any issues and the Google Maps Platform team responding quickly to resolve them. "Google Maps Platform and Searce helped us understand the depth of the platform and, through daily and weekly standups, reduced the proof of concept cycle time and enabled us to start the journey of making it a core part of our system," says Katta.

"Google has
been with us
throughout this
journey and its
team cares
about how its
platform enables
our business.

The team's
experiences and
the lessons
we've all learned
through this
process are
being fed back
into the platform

to further
develop its
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Refining package delivery and car sharing services

With Mobility services integrated into its mobile app, Rapido is looking to better combine package deliveries with customer rides and expand a car sharing service that already operates in tier one cities.

"We're looking at how we better combine and batch customer journeys with package deliveries in order to improve Captains' earnings while helping them control their fuel costs," explains Katta.

Further, Rapido is looking to expand its shared car offering. "The biggest challenge with sharing is to ensure the mapping is correct," says Katta. "The

permutations of picking up and dropping off multiple people to different destinations become very complex, and this is where we are betting on Mobility services to build out this experience. We have a very good shot at getting this right because we have the right technology and the right partners to help us."

Katta commends the support that the business received from the Google Maps Platform team during the project. "Google Maps Platform has been with us throughout this journey and its team cares about how its platform enables our business," he says. "The team's experiences and the lessons we've all learned through this process are being fed back into the platform to further develop its features and services. Overall, the relationship is helping us take this important journey much faster."